

AYUSH LODH

Curriculum Vitae

Shastrinagar, Burnpur
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Work Experience

- Feb, 2024 – **Research Intern**, *Computer Vision and Pattern Recognition Unit*, Indian Statistical Institute, Kolkata, India.
April, 2024 Worked on online and offline handwritten text recognition from a webcam or camera live feed.
- May, 2024 – **Junior AI Engineer**, *Habitat Lens Private Limited*, Kolkata, India, AI Solutions.
July, 2024 Gained comprehensive exposure to Handwritten Document Images, Annotation Task
- Dec, 2025 – **Research Intern**, *Computer Vision Center*, Universitat Autònoma de Barcelona, Spain (Remote).
Mar, 2026 Worked on Multimodal RAG using Graph Neural Networks on Enterprise Document Images

Education

- 2025 – **Master of Technology, Computer Science Engineering (Analytics)**, *National Institute of Technology*, Delhi, India. CGPA: 7.5/10.
Present **Courses:** Data Mining and Warehousing, Deep Learning, Advanced Databases
Ongoing Project: From Chunks to Graphs, *Advisors: Prof. Dr. Nisha Singh Chauhan, NIT Delhi & Dr. Sanket Biswas, CVC, UAB*
- 2020 – 2024 **Bachelor of Engineering, Information Technology**, *University Institute of Technology*, Burdwan University, India. CGPA: 8.79/10.
Courses: Image Processing, Artificial Intelligence, Machine Learning
Final Year Project: 3D - Playfair Cipher with Dual Key, *Advisor: Ms. Nabanita Das*

Certifications

- Jan – Apr 2022 **Problem Solving through Programming in C**, *Translate basic algorithms to programs (in C language) and apply programming to solve matrix addition and multiplication problems and searching and sorting problems*, IIT Kharagpur, NPTEL.
- Jan – Apr 2022 **An Introduction to Programming through C++**, *Organizing large programs into units such as functions and classes and implemented standard libraries of C++ like string, vector and map classes.*, IIT Bombay, NPTEL.
- Jul – Sept 2023 **Programming, Data Structures and Algorithms Using Python**, *Translate basic algorithms to programs (in Python language) and apply programming to understand and implement different data structures to solve problems like searching and sorting problems. Organizing large programs into units such as functions and classes. Implemented different OOP, error handling, recursions, backtracking and dynamic programming concepts*, IIT Madras, NPTEL.
- Jul – Oct 2023 **Machine Learning and Deep Learning - Fundamentals and Applications**, *Studied traditional Machine Learning approaches, e.g. Bayesian Classification, Multilayer Perceptron etc. and modern Deep Learning architectures like Convolutional Neural Networks, Autoencoders etc.*, IIT Guwahati, NPTEL.

Publications

In Conference Proceedings

- 2025 **Ayush Lodh**, Ritabrata Chakraborty, Shivakumara Palaiahnakote, and Umapada Pal. A transformer based handwriting recognition system jointly using online and offline features. In *Asian Conference on Pattern Recognition*, pages 250–264. Springer, 2025.
- 2025 Aniket Pal, Sanket Biswas, Alloy Das, **Ayush Lodh**, Priyanka Banerjee, Soumitri Chattopadhyay, Ajoy Mondal, Dimosthenis Karatzas, Josep Lladós, and CV Jawahar. Icdar 2025 handwritten notes understanding challenge. In *International Conference on Document Analysis and Recognition*, pages 553–567. Springer, 2025.

Preprint

- 2025 Aniket Pal, Sanket Biswas, Alloy Das, **Ayush Lodh**, Priyanka Banerjee, Soumitri Chattopadhyay, Dimosthenis Karatzas, Josep Lladós, and CV Jawahar. Bank: Benchmarking neural transcription and search for scientific notes understanding. *arXiv preprint arXiv:2504.09249*, 2025.

Research Experience

Indian Statistical Institute, Kolkata, India.

- Feb, 2024 – **A Transformer Based Handwriting Recognition System Jointly Using Online and Offline**
April, 2024 **Features**, Advisor: Prof. Dr. Umapada Pal.

Proposed an end-to-end transformer for writer-independent handwriting recognition that fuses offline image patches and online stroke tokens via learnable latent queries in a shared space.

Habitat Lens Private Limited, Kolkata, India.

- May 2024 – **NoTeS-Bank: Benchmarking Neural Transcription and Search for Scientific Notes Un-**
July 2024 **derstanding**, Advisor: Mr. Alloy Das.

Introduced a benchmark for transcription and search over scientific handwritten notes with mixed equations, diagrams, and text, defining Evidence-Based and Open-Domain VQA tasks for vision–language models.

Universitat Autònoma de Barcelona, Spain

- Dec, 2025 – **From Chunks to Graphs**, Advisors: Dr. Sanket Biswas & Dr. Josep Lladós.

Mar, 2026 Developed a training-free multimodal retrieval framework replacing chunk-based selection with structure-aware subgraph retrieval over Doc2Graph representations, improving factual consistency on document VQA over text-RAG baselines.

Skills

- Programming Python, C/C++
Libraries PyTorch, Tensorflow
Documentation MS Office, L^AT_EX
Languages English, Bengali (Native) & Hindi